WORKED SOLUTION

STEP I 2006, Question 2

Pure • Areas and Optimisation

STEP I 2006, Question 2

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/06-s1-q2

The key idea

Fix the tether point by a single number, its distance x from one corner, and follow the rope as it wraps around the barn. The grazing region then splits into a semicircle and four quarter-circles whose radii are simple linear functions of x. So A is a quadratic in x on a closed interval, and both parts of the question come down to finding its largest and smallest values.

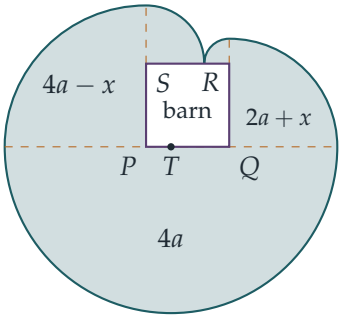
Solution

Setting up

Label the barn $PQRS$, with the goat tethered at T on side PQ, where $PT = x$ and $TQ = 2a - x$ for $0 \leq x \leq 2a$. By symmetry it doesn't matter which side the goat is on.

The goat can reach a point when the shortest path from T to it that avoids the barn is at most $4a$. Taut, that path is straight except where it bends around the corners of the barn. Going round the barn from T:

- **In front of PQ.** Every point on the far side of the line PQ can be seen directly from T, so the goat grazes a semicircle of radius $4a$.
- **Beside PS and QR.** Once the rope bends round P, $4a - x$ is left, and it sweeps a quarter-circle centred at P alongside side PS. Likewise, $4a - (2a - x) = 2a + x$ is left at Q, giving a quarter-circle centred at Q alongside QR.
- **Behind the barn.** Since $x \leq 2a$, we have $4a - x \geq 2a$, so the rope reaches S with $(4a - x) - 2a = 2a - x$ to spare, and that sweeps a quarter-circle centred at S behind side SR. In the same way, the rope reaches R with $(2a + x) - 2a = x$ left over, giving a quarter-circle of radius x centred at R.



Ask yourself Could the two pieces behind the barn overlap, so that we count some grass twice?

The quarter-circles centred at S and R lie on the same side of the line SR. Their radii, $2a - x$ and x, add up to $2a = SR$, so the two circles just touch at the point of SR that is $2a - x$ from S. So they don't overlap, and between them the goat reaches the whole of side SR. No other pieces overlap: the semicircle lies on one side of the line PQ and the four quarter-circles lie on the other, each in its own region beside or behind the barn. And a point behind the barn is reachable exactly when it lies within the remaining rope of S or of R, so nothing is missed.

The area

Adding the semicircle and the four quarter-circles,

$$\begin{aligned} A &= \tfrac{1}{2}\pi(4a)^2 + \tfrac{1}{4}\pi\left[(4a-x)^2 + (2a+x)^2 + (2a-x)^2 + x^2\right] \\ &= 8\pi a^2 + \tfrac{1}{4}\pi\left(4x^2 - 8ax + 24a^2\right) \\ &= \pi\left(x^2 - 2ax + 14a^2\right) \\ &= \pi\left[(x-a)^2 + 13a^2\right]. \end{aligned}$$

This formula holds for every x in $[0, 2a]$. At the ends ($x = 0$ or $x = 2a$, goat tied at a corner) one quarter-circle shrinks to a point, and the semicircle and the quarter-circle of radius $4a$ join up into three-quarters of a circle, which is what we would expect.

The maximum and the minimum

On $0 \le x \le 2a$ we have $-a \le x - a \le a$, so $0 \le (x-a)^2 \le a^2$. Hence

$$13\pi a^2 \le A \le 14\pi a^2,$$

which proves $A \le 14\pi a^2$. Equality on the right holds exactly when $x = 0$ or $x = 2a$, that is, when the goat is tied at a corner. The left-hand bound is attained when $x = a$, so the minimum value of A is

$$A_{\min} = 13\pi a^2,$$

reached when the goat is tethered at the midpoint of a side.

Checks

- At a corner ($x = 0$), we get three-quarters of a circle of radius $4a$ ($12\pi a^2$), a quarter-circle of radius $2a$ round the far corner of the adjacent side (πa^2), and a quarter-circle of radius $2a$ behind the barn (πa^2). That makes $14\pi a^2$, as the formula says.
- We estimated the grazed area on a fine grid with $a = 1$, testing each point by its true shortest distance round the barn (a straight line, or one bent at the corners). We got 40.845 at the midpoint against $13\pi \approx 40.841$, and 43.988 at a corner against $14\pi \approx 43.982$, with similar agreement at other tether points.

Common slips

- Forgetting the pieces behind the barn: the rope is long enough to go round two corners on each side.
- Worrying that the two pieces behind the barn overlap, or forcing an overlap into the calculation. Their radii always add up to exactly $2a$, so the circles only touch.

Beyond STEP

The taut rope follows a *geodesic*, the shortest path in a region with an obstacle. Here it is made of straight segments that bend only at the corners of the barn, so the set the goat can reach is a "ball" in a non-Euclidean way of measuring distance. Shortest paths around obstacles lead to the calculus of variations and to motion-planning problems in computer science. Siklos, *Advanced Problems in Mathematics*, has more problems of this kind, where a careful case analysis turns a geometry question into simple algebra.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.